SentraCore - Group 3
CSIT321 - High Level Design Document

By: Sara Ranjbar (7817848), Sikha Nair (7810398), Sharen Thomas (7700787), Yaxin Wang (7741091) & Cadence D'Souza (7769714)

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Introduction

This High-Level Design Document provides a comprehensive overview of the SkillSync platform, a web-based collaborative learning and career readiness system. It provides a structured and accessible overview of the system's core features, architecture, and intended functionality. The content is designed to be easily understood by both technical and non-technical audiences.

The document outlines key requirements, assumptions, architectural plans, application modules, and user flows. It also covers user interface design, deployment environments, security considerations, and identified risks along with proposed mitigation strategies. Visual elements such as diagrams are included to support understanding of system behavior and structure.

As part of the development process, this document serves as a foundational reference to guide implementation and ensure alignment across the project team. It also helps stakeholders and reviewers assess whether the system's design meets the intended goals and expectations.

Requirement Summary

Functional Requirements:

- **User accounts and roles:** The system should let users sign up, log in, and create profiles. There will be different types of users like regular learners, mentors, and admins. Each type of user will have different things they can do. Mentors can show when they're available to help, and learners can find help or join study groups.
- **Study pods**: users can create or join study pods based on common goals. Like passing a test or learning a skill. Each pod lets users track their progress, use a shared whiteboard, and write notes. Users can choose the mode on how the pod runs such as group or individual or hybrid.
- **Peer link:** Enables real-time peer-to-peer support via a topic-based question board, live chat, and optional scheduled help sessions. The Study Buddy Match tagging lets users finds responders to their problem based on learning type, skills and so on.
- **AI Interview Practice:** Users can simulate job interviews using AI avatars tailored to specific job roles and interviewer personas. The system provides instant AI-driven

feedback on performance metrics such as tone, clarity, and confidence, allowing learners to improve through iterative practice.

- **Rating and Feedback:** After sessions, users can rate each other based on how helpful and clear they were. These ratings show up on profiles and help build trust in the community. High scores get special tags like "Top Mentor."

Non-Functional Requirements:

- **Usability:** The interface should be intuitive, responsive, and optimized for both desktop and mobile use. The system must guide users seamlessly through features such as pod creation, buddy matching, and avatar interview sessions.
- **Availability and scalability:** The platform should be available 24/7 with minimal downtime. It should support scaling to accommodate growing numbers of concurrent users, especially during peak academic seasons or career events.
- **Security and privacy:** All user data must be securely stored and transmitted using encryption protocols. Access control mechanisms must prevent unauthorized access. Personally identifiable information and session data (e.g., notes, recordings) should comply with privacy standards.
- **Fast performance:** Pages and actions should load quickly. For example, chats and feedback should appear without long delays.
- **Network requirements:** Requires stable internet access for real-time collaboration features such as Quick Help Live Chat, whiteboard interaction, and AI interview simulations. Offline functionality for note viewing/editing is optional but beneficial.
- **System integration:** SkillSync should connect well with tools like cloud storage (for saving files) and AI services (for interview feedback).
- **Reliability and fault tolerance:** The system must handle server or module failures gracefully, with fallback mechanisms such as autosave for notes and retry attempts for interrupted interview sessions.

Assumptions and Prerequisites

Assumptions:

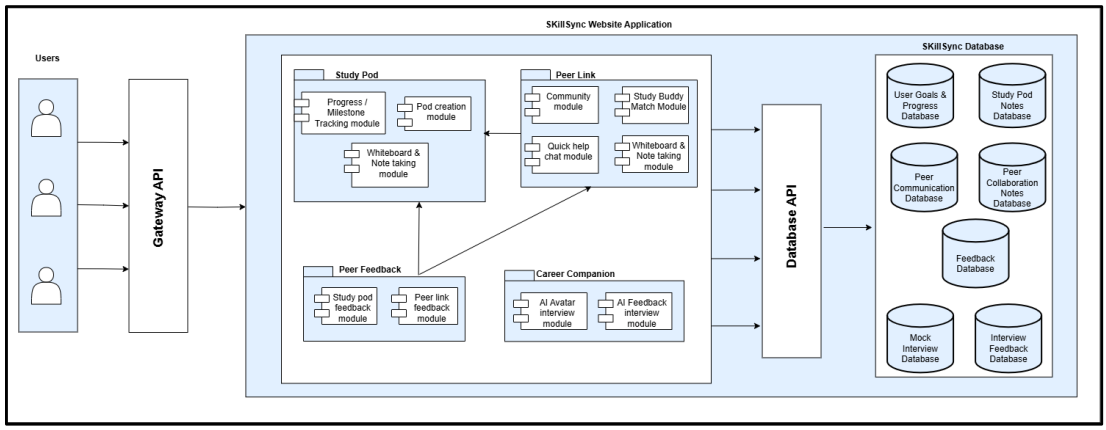
- Users engage in collaboration, feedback, and regular activity.
- Users join/create pods based on topic and timeline fit.
- Users are expected to set goals, track progress, and check in.
- Pods support online collaboration modes.
- Users help peers via PeerLink and Quick Help Chat.
- Users agree to AI feedback like mock interviews and tone analysis.
- Peer ratings help ensure collaboration quality and trust.

Prerequisites:

- Core matching algorithms for Study Buddy Match.
- AI avatar-based feedback (e.g., roleplay interviewer) must be integrated with LLMs.
- Shared whiteboards and live communication tools must be embedded, not recreated.
- Visual progress tracking and journal entries must be light and easy to use.
- Calendar reminders, task checklists, and the dashboard must be modularly implemented.
- All interactions (pods, help, feedback) are tracked back into a central user profile and reputation system.

High-Level Design

- **High level Software Design Diagram**



Users can interact with the website application of SkillSync through several learning, communication, and cooperation and career related services. The users will be connected to the application through a gateway REST API which connects the

interface to the system as well as retrieves and amends data by connecting the frontend and backend..

SkillSync Web Application has 4 main areas:

- **Study Pod**: For collaborative goal-based learning.
    - Create pods, track progress and milestones, and take notes on the shared whiteboard.

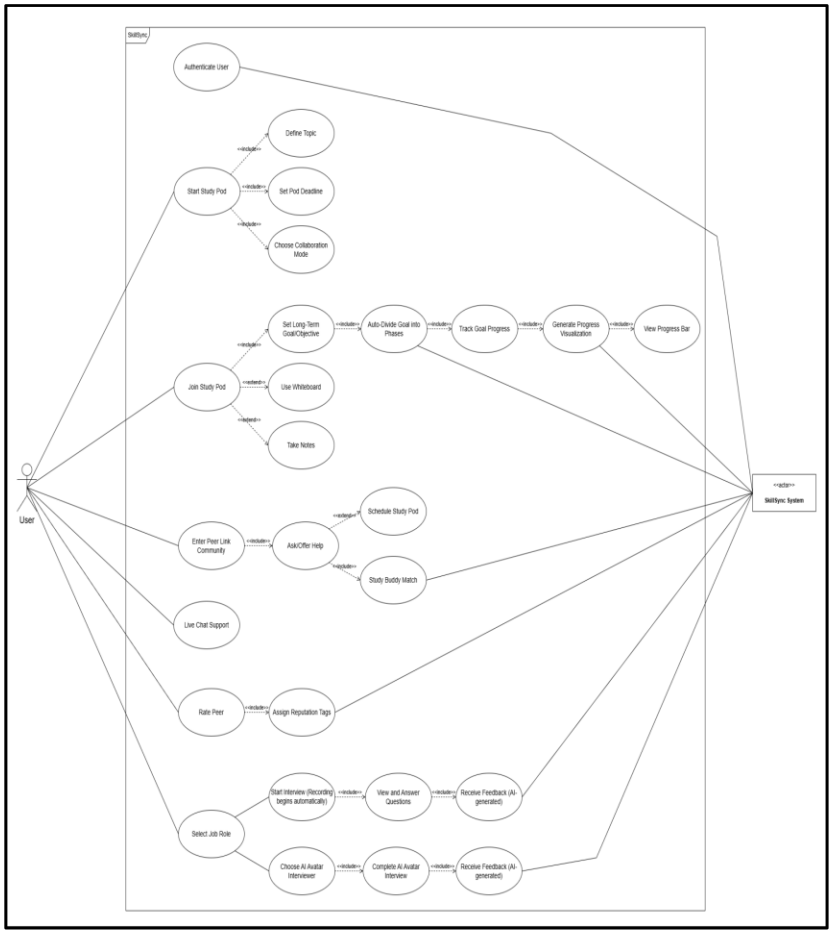
- **Peer Link**: For peer learning and support.
    - Community and Study Buddy Matching
    - Quick Help Chat, Shared Whiteboard, and Note-Taking

- **Peer Feedback**: For systematic peer reviews.
    - Feedback from Study Pod and Peer Link sessions

- **Career Companion**: For Career Readiness
    - AI Avatar Interviews (Avatar Based)
    - AI evaluations of interview performance.

The **Database** of SkillSync manages user data such as objectives, notes, interaction, teamwork, comments, and practice interviews.

- **Use Case Diagram**



Use Case Categories and Descriptions:

1. **Actors:**

   Primary Actor

   User: The primary actor who interacts with the system.

   Secondary Actor

   SkillSync System: It works alongside the User to carry out specific tasks.

2. **Use Cases (Ovals):**

   Use cases **connected to User** (specific actions a user can perform):

   - Start Study Pod
   - Join Study Pod
   - Peer Link Community
   - Live Chat Support
   - Rate Peer

- Select Job Role

Use cases **connected to SkillSync System** (performed by the system):

- Authenticate
- Auto-Divide Goal into Phases
- Generate Progress Visualization
- Study Buddy Match
- Assign Reputation Tags
- Receive Feedback (AI-generated)

3. **<<include>> and <<extend>> Relationships:**
   The **<<include>> relationship** indicates that a use case **always incorporates** the behavior of another use case as part of its process. The included use case is executed every time the base use case runs, and it cannot function independently.

   The **<<extend>> relationship** indicates that a use case **optionally adds behavior** to another use case under certain conditions. The extended use case depends on the base use case but is not always executed.

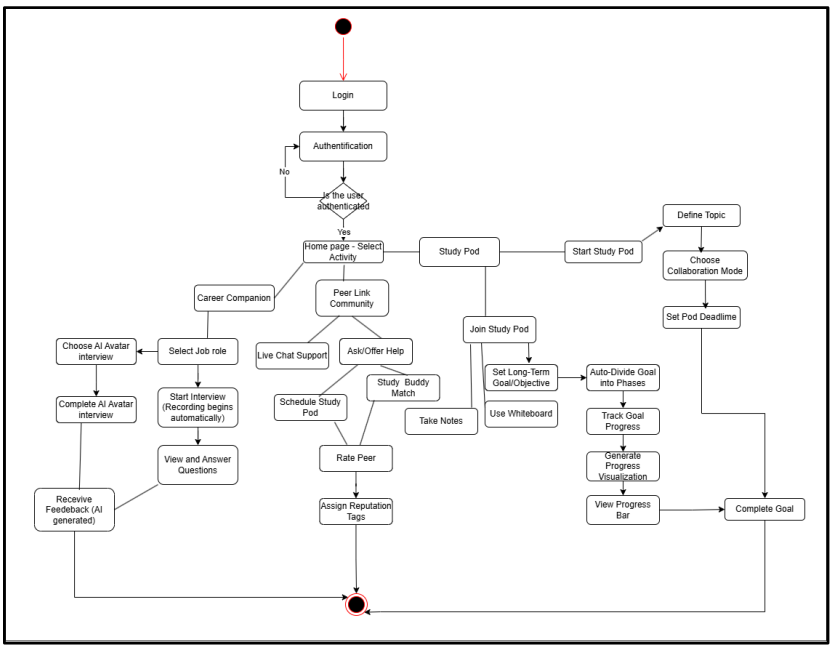
   - When the user performs the **"Start Study Pod"** use case, they will also **define the topic**, **set the pod deadline**, and **choose the collaboration mode** as part of the overall workflow. These actions are mandatory and are therefore modeled using **<<include>> relationships**.
   - When the user performs the **"Join Study Pod"** use case, they will **set a long-term goal or objective**, **track their goal progress**, and **view a visual progress bar** as required steps. Additionally, they can optionally **use the whiteboard** and **take notes** during the session.
   - When the user performs the **"Enter Peer Link Community"** use case, they will **ask for or offer help** and use the **study buddy match**, which is modeled as an **<<include>> relationship**. They may also optionally **schedule a Study Pod** to continue the discussion in more depth. This optional action is modeled using an **<<extend>> relationship**.
   - When the user performs the **"Rate Peer"** use case, the system will automatically **assign reputation tags** based on its average calculation.

This is a required outcome and is modeled using an **<<include>> relationship**.

- When the user performs the **"Select Job Role"** use case, they must then choose between two distinct interview methods: **Start the interview (recording begins automatically)** or **choose an AI avatar interviewer**.

  If the user starts the interview, they will **view and answer questions**, then **receive AI-generated feedback**. If they choose an avatar interviewer, they will complete **the avatar-led interview** and then receive **AI-generated feedback**. These actions are mandatory within each selected path and are therefore modeled using **<<include>> relationships**.

- **Activity Diagram**



- **Login and Authentication:**
  - The user starts by logging in.
  - If authentication fails, the user is redirected and asked to try again.
  - If successful, they access the homepage to choose an activity.

- **Study Pod:**
  - Users can either **start** or **join** a Study Pod.

    - To start one, they define the topic, choose collaboration mode, and set a deadline.
    - They then set long-term goals, which the system breaks into manageable phases.
    - Users can collaborate using whiteboards, take notes, and track progress through a visual bar and progress dashboard.
    - Once goals are completed, the pod session is marked as complete.

- **PeerLink Community:**
    - This module enables collaborative help:
    - Users can **ask or offer help**, **schedule pods**, or use **live chat support**.
    - The **Study Buddy Match** feature connects users with responders such as mentors or peers who would help solve the user's problem with suitable learning type, skills, resources and so on.
    - After sessions or help, users **rate their peers**, and the system assigns **reputation tags** based on feedback.

- **Career Companion:**
    - **Practice with AI Avatar**
        - The system launches a virtual mock interview with a realistic **AI avatar**.
        - Users interact with the avatar by answering job-related questions.
        - After completing the session, the AI automatically evaluates the user's performance and gives **feedback** on tone, clarity, and confidence.

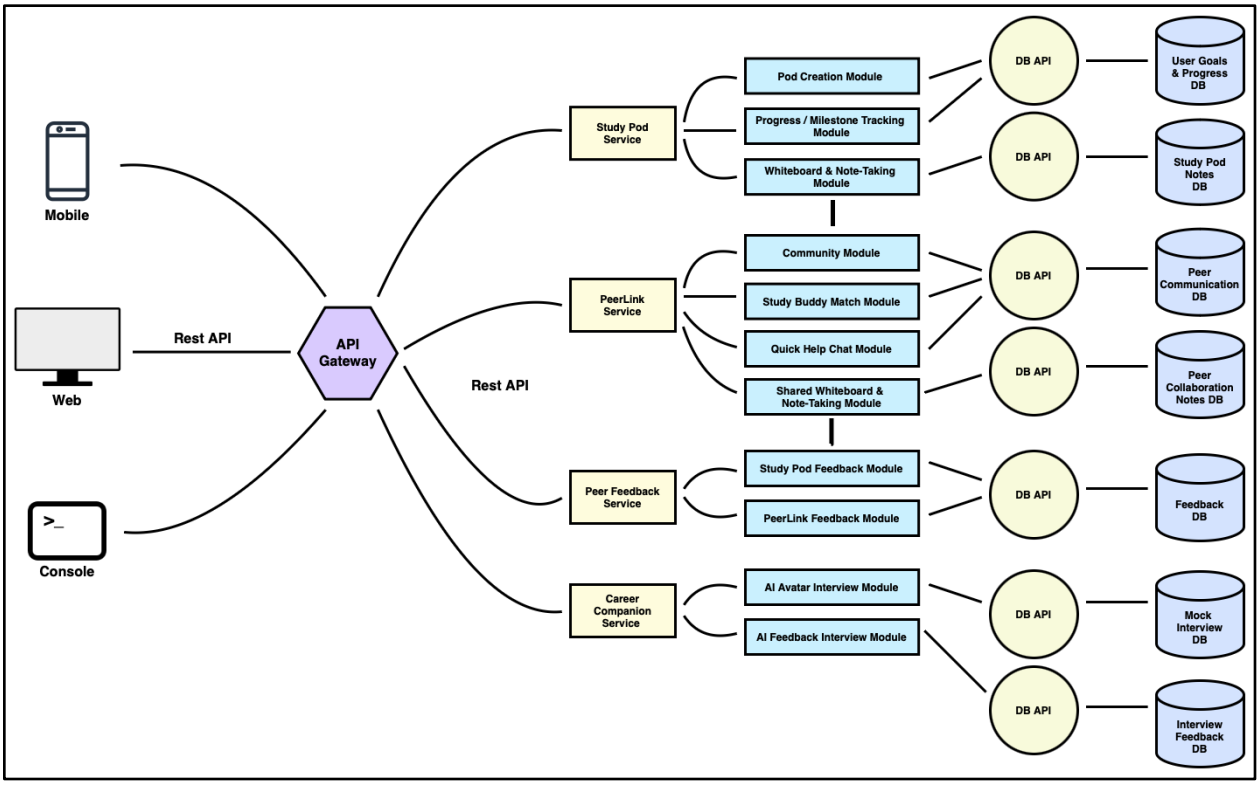
    - **Self-Recording Mode**
        - The system automatically starts **recording** when the user begins.
        - Users answer the displayed questions.
        - Once the session ends, the recording is submitted.
        - The system processes the video and delivers **AI-generated feedback** on how the user performed.

- **Integration Issues**

  Many modern tools, including React, Node.js, MongoDB, Firebase, OpenAI, and WebRTC, are integrated by SkillSync+; each offers a distinct advantage but also raises possible integration issues. When third-party APIs, such as OpenAI or Firebase, encounter rate limitations, outages, or modifications to their answer formats, problems could occur. Similar to this, unexpected errors or failures may result from structural mismatches with the database or from inconsistencies between both backend and frontend data structures. Features that depend on media handling, such live peer support sessions and AI avatar interviews, might also be impacted by browser bugs (e.g., microphone access via WebRTC). Although these hazards can be proactively controlled, they are innate to multi-service systems.

Architectural Plan

- **Physical Architecture**



  - The physical architecture design of SkillSync platform demonstrates how users communicate with different backend services via a modular and unified system framework.

- On the left, the client interfaces such as mobile, web and console (CLI) are shown as the primary access entry points. These frontend clients interact with the system through REST API calls, enabling the users to request services such as engaging in peer discussions, joining study pods, or doing mock interview sessions.

- All the incoming user requests are forwarded to the API Gateway, which works as a main controller. It is responsible for handling the distribution of the requests to the correct backend systems, keeping a track of user session management, applying access control, and maintaining a secure and smooth communication flow between the server infrastructure and client interface.

- The API Gateway distributes each incoming requests to one of the four main functional service modules:
  - ➔ Study Pod Service, which handles the creation of study pods, milestone tracking, and facilitates note-taking.
  - ➔ Peer Link Service, which handles peer matching, chat and collaboration tools.
  - ➔ Peer Feedback Service, which handles the ratings and user feedback based on their interactions.
  - ➔ Career Companion Service, which handles AI-driven mock interview sessions and delivers feedback.

- Every service contains several functional modules. For instance, the Study Pod Service includes features such as creation of study pods, tracking progress, and collaborative whiteboard and note-taking. Likewise, the PeerLink Service includes features such as study buddy match, community discussions, quick help chat and shared whiteboard and note taking.

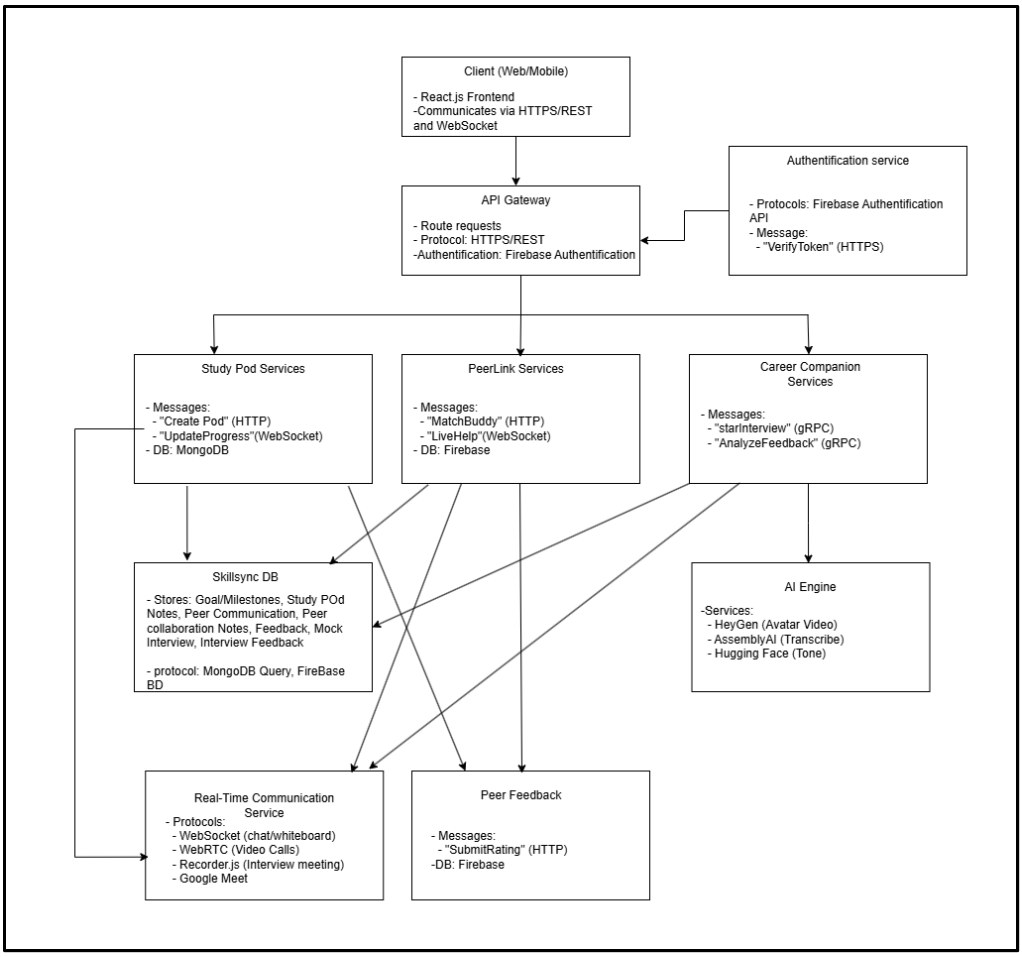
- These modules communicate via a DB API (Database API), which functions as an interface between the system's processing logic and its respective database. The DB API handles the database queries, validates the data, enforces business rules, and guarantees protected access to stored information.

- On the far right, every DB API is linked to a specific assigned database, which safely stores the data associated with its relevant service. It includes User Goals & Progress DB, Study Pod Notes DB, Peer Communication DB, Peer

Collaboration Notes DB, Feedback DB, Mock Interview DB and Interview Feedback DB.

- This structure ensures that every service is individually scalable, modular and well-protected, while the use of dedicated DB APIs promotes ease of maintenance and logical separation. The architecture is designed to enable live collaboration, flexible deployment, and smooth integration of future enhancement with minimal disruption.

- **Logical Architecture**



Client (Web/Mobile)

This is the main app interface that is used by users, and it is built using React.js. It is mobile and browser compatible, and it talks to and from the data using

HTTP/REST APIs and WebSocket. The client, users can create or join, ask for help, give feedback and prepare for interviews.

API Gateway
The API Gateway acts like a traffic controller. It receives all of the client requests and figures out where they should go. It checks if the user is authenticated with Firebase Authentication and routes the request to the correct service like Study Pods, Peer Help, or Interview Tools. It supports both the normal HTTP and real-time WebSocket messages.

Authentication Service
It makes sure that only authenticated and registered users have access to the site. It also integrates Firebase Authentication for the verification of login credentials. It uses a "VerifyToken" function in order to verify a user's token before granting access to anything.

Study Pod Services
This controls everything regarding Study Pods. Users can start pods by setting a goal, deadline, and subject, or join pods. It controls two actions: creating a pod via HTTP and sending in-progress progress via WebSocket. It saves pod details like goals and member information in a MongoDB database. It is integrated with other systems for in-progress monitoring and communication.

PeerLink Services
It is peer-to-peer support, where the users can either ask for help or offer it. There are two main features: finding a study partner to study with and asking for immediate help. Matching is done using HTTP, and live help sessions using WebSocket. Everything user-related and help sessions are saved on Firebase.

Career Companion Services
This service helps users prepare for job interviews. It uses gRPC to start mock interviews and go through feedback. It gets linked to the AI Engine for simulating interviewers and offering intelligent feedback. Users are able to practice speaking, get guided, and get answers polished with AI tools.

SkillSync DB
This is the primary storage of user progress and collaboration information. It holds

all the goals, milestones, peer messages, notes, feedback, mock interviews, etc. It uses both MongoDB and Firebase databases. All other services forward their information here; hence, it is the core brain of the platform.

Real-Time Communication Service
This powers all live functionality like chat, whiteboard, and video calls. It uses WebSocket for text and whiteboard, and WebRTC for video. There is even a choice of using Google Meet for meetings. It includes a recorder to record mock interview sessions. This enables users to work harmoniously together in real time.

Peer Feedback
The users can leave ratings and comments on each other after each session. It builds a system of trust with rewards for helpfulness to the users. It supports a 5-star rating system and stores the data in Firebase. The users are rewarded with tags like "Top Mentor" or "Trusted Contributor" depending upon their average score.

AI Engine
This part adds smart AI features to the platform. It includes:

- HeyGen for interview video avatars.
- AssemblyAI for speech-to-text.
- Hugging Face for tone and emotional check-in responses.

These assist users to improve the way they speak, sound, and behave when doing imitation interviews or group work.

Application Modules

- **Core components**
    - **Study Pods**: This feature facilitates learning and goal oriented tasks in one place. Mentors can start classes and users can join to learn and challenge themselves. Users can also create a pod with different learning objectives for example an exam prep pod, making it a single user or multi user pod according to the user's preference. To make learning easier the system gets the goal input from the user and creates a progress tracking/ milestone pathway for the user to follow. These pods also integrate whiteboards to enhance learning visually.
    - **PeerLink**: It is built to promote a collaborative educational environment by enabling users to either receive help or by contributing to offer guidance on academic subjects. This feature facilitates community-based learning and also

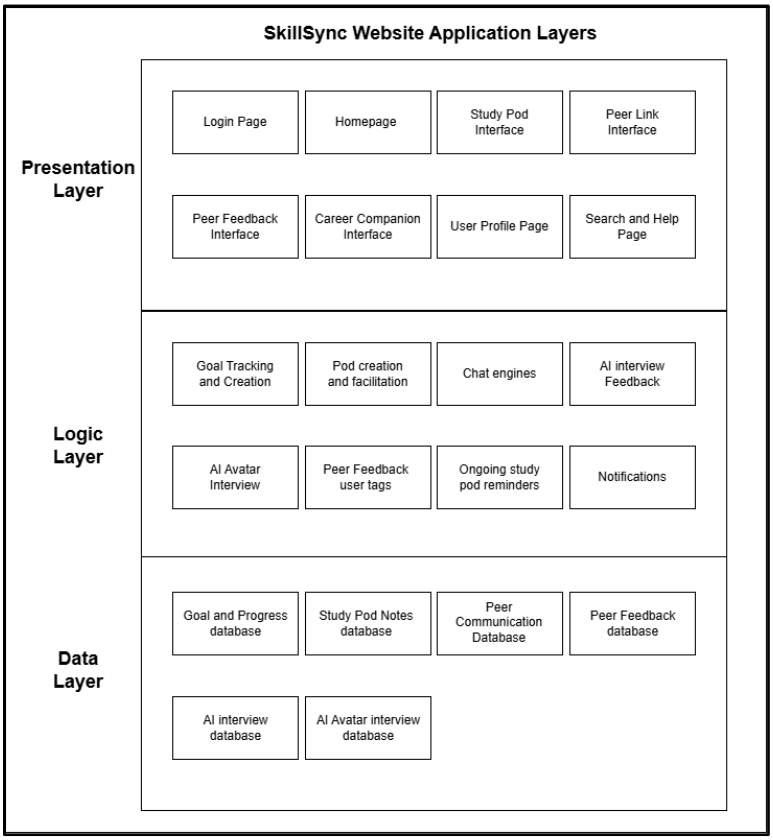
accommodates both flexible and scheduled learning engagements. The key functional sub-components consist of: **Community Q&A Mechanism,** where the learners can raise questions or topics to discuss, which other community members can give responses. **Study Buddy Match,** a smart system which links users with common interests, learning objectives or academic backgrounds. **Quick Help Live Chat, f**or individuals who are seeking instant support, this messaging feature delivers immediate, real-time guidance. **Mentor Status Display,** where the mentors can mark their status to "Available to help", which is visible to all the users needing guidance. **Collaborative Whiteboard Tool,** where PeerLink offers a shared space where the users can map out ideas, sketch or work through problems together. **Integration of Study Pod Meetings,** when detailed discussions are necessary, users can initiate a session via Study Pod system right from PeerLink. These sessions support screen-sharing options for a better demonstration, while keeping the camera disabled to help users remain comfortable and promote a focused setting.

- **Peer Feedback System:** This system enables the users to review each other after each interaction session, supporting an evolving reputation network within the application to guarantee accountability, continuous improvement and quality of sessions. The key functional sub-components consist of: **5-Star Rating System, f**ollowing a session via PeerLink or Study Pods, the users can rate the other person they worked with on factors such as communication, preparedness and explanation clarity. **Profile Reputation Indicator, where e**very user's account has a publicly viewable rating displayed on their profile, an aggregated score is calculated from all the feedback that is received. **Peer Reputation Computation, where t**he system analyzes the collected feedback to produce a Reputation Score that reflects the quality and consistency of the user's participation over time. **Recognition Tag Mechanism, a**ccording to rating thresholds, users are given performance based tags such as "Top Mentor" for rating levels between 4.8 to 5.0. These tags appear on their profiles which serves as an encouragement and also helps new users as a trust indicator.

- **Career Companion:** In this feature users have the option to practice their interview with 2 types of interview feature. First being a simple feature where the user gets asked a question(s) by the system which once it's over will be analyzed for improvement for the user. Second is where there are AI avatars

with different personalities to give user's the feel of an actual interview setup which they will later on face, and once it's over, it will be again analyzed, and feedback will be shared to the user.

- **User interfaces and layers**



The SkillSync website follows a three-tier architecture consisting of the Presentation Layer, Logic Layer, and Data Layer. This structure helps keep the website efficient, adaptable, and manageable. Here's a breakdown of each layer and its components:

1. **Presentation Layer**
   This is the **user interface layer**, what users directly interact with when using the SkillSync platform. It includes:
   - **Login Page:** Entry point for user authentication.
   - **Homepage:** Central landing page after login, possibly showcasing navigation, announcements, or updates.
   - **Study Pod Interface:** UI to join or manage collaborative study pods with shared goals.

- **Peer Link Interface:** Interface for connecting and collaborating with peers.
- **Peer Feedback Interface:** Allows users to view or give feedback on peer performance.
- **Career Companion Interface:** Interface for interview preparation.
- **User Profile Page:** Personal user dashboard displaying personal details and activity.
- **Search and Help Page**: Interface to search for information or access user support.

2. **Logic Layer**
   This layer contains the **core functionalities and business logic** that power the platform's features. It processes inputs from the Presentation Layer and communicates with the Data Layer. Components include:
   - **Goal Tracking and Creation:** Manages goal setting for study pods.
   - **Pod Creation and Facilitation:** Logic to create, manage, and facilitate group study pods.
   - **Chat Engines:** Handles real-time messaging and interaction between users.
   - **AI Interview Feedback:** AI system to provide suggestions and improvements for mock interviews.
   - **AI Avatar Interview:** Simulates interview scenarios using AI-driven avatars.
   - **Peer Feedback User Tags:** Handles tagging or categorization of users based on feedback.
   - **Ongoing Study Pod Reminders:** Sends reminders for deadlines and group sessions.
   - **Notifications:** System-generated alerts for user activity or updates.

3. **Data Layer**
   This is the **storage layer**, containing structured data repositories that support the logic and presentation layers. It ensures data integrity, security, and efficient access. Key databases include:
   - **Goal and Progress Database:** Stores information on user goals and tracking metrics.
   - **Study Pod Notes Database:** Saves notes within study pods.

- **Peer Communication Database:** Logs chats and peer interactions for reference and analytics.
- **Peer Feedback Database:** Archives all feedback given and received between users.
- **AI Interview Database:** Stores user attempts and results of AI-generated interview sessions.
- **AI Avatar Interview Database:** Contains data related to the simulated avatar interview interactions.

- **The database layer**

The SkillSync platform employs both MongoDB and Firebase storage as its primary data handling solutions. This multi-layered approach enables both structured data (i.e., user objectives and chat history), and unstructured data (i.e., interview recordings) are handled with safety and efficiency.

MongoDB functions as the system's core NoSQL database. It is especially ideal for SkillSync due to its adaptable schema, which enables dynamic adjustments and seamless integration across various modules. This is crucial for managing diverse and expanding datasets such as learning progress, collaborative contents, and peer evaluations. MongoDB facilitates real-time data access and enables smooth integration with backend systems created in Node.js.

Firebase Storage is utilized to handle binary data and media files. This encompasses assets such as video and audio recordings from interview recordings, avatar-based feedback sessions, and captured whiteboard screenshots. Firebase provides robust security rules for access of storage, guaranteeing only users are permitted for uploading and retrieval of confidential data. Its built-in ability ensures seamless performance under intense workloads such as multiple video uploads or simultaneous feedback retrieval.

For efficient data management, the platform arranges its backend into well-defined logical collections or databases that correspond to particular use cases across the application:

- **Peer Communication Database:** This database is designed to store all real-time message-based conversations between the users on SkillSync. It gathers

messages from Quick Help Chat, public forums, and direct peer interactions, including the timestamps and sender/recipient details. This preserves the conversations for future access, supervision or review. It also helps to sustain the ongoing learning by enabling users to revisit past conversations. In total, it supports a responsive and structured communication environment across the platform.

- **Peer Collaboration Notes Database:** This database stores all written and visual contents that are created during the learning sessions. This encompasses shared whiteboard drawings, jointly generated diagrams, brainstorming drafts, and any team-based or individual materials created during PeerLink or Study Pods meetings. These resources are stored with metadata of sessions and timestamps for quick access. This database enables users the ability to revisit and build on previous collaborative efforts, which makes it particularly useful for long-duration projects and those who learn asynchronously. It acts as the visual record of SkillSync's team-oriented features.

- **User Goals and Progress Database:** This database stores the goals users set inside Study Pods, along with their progress. It tracks completed milestones, updates progress bars, and helps users stay on track by showing what they've finished and what's left to do.

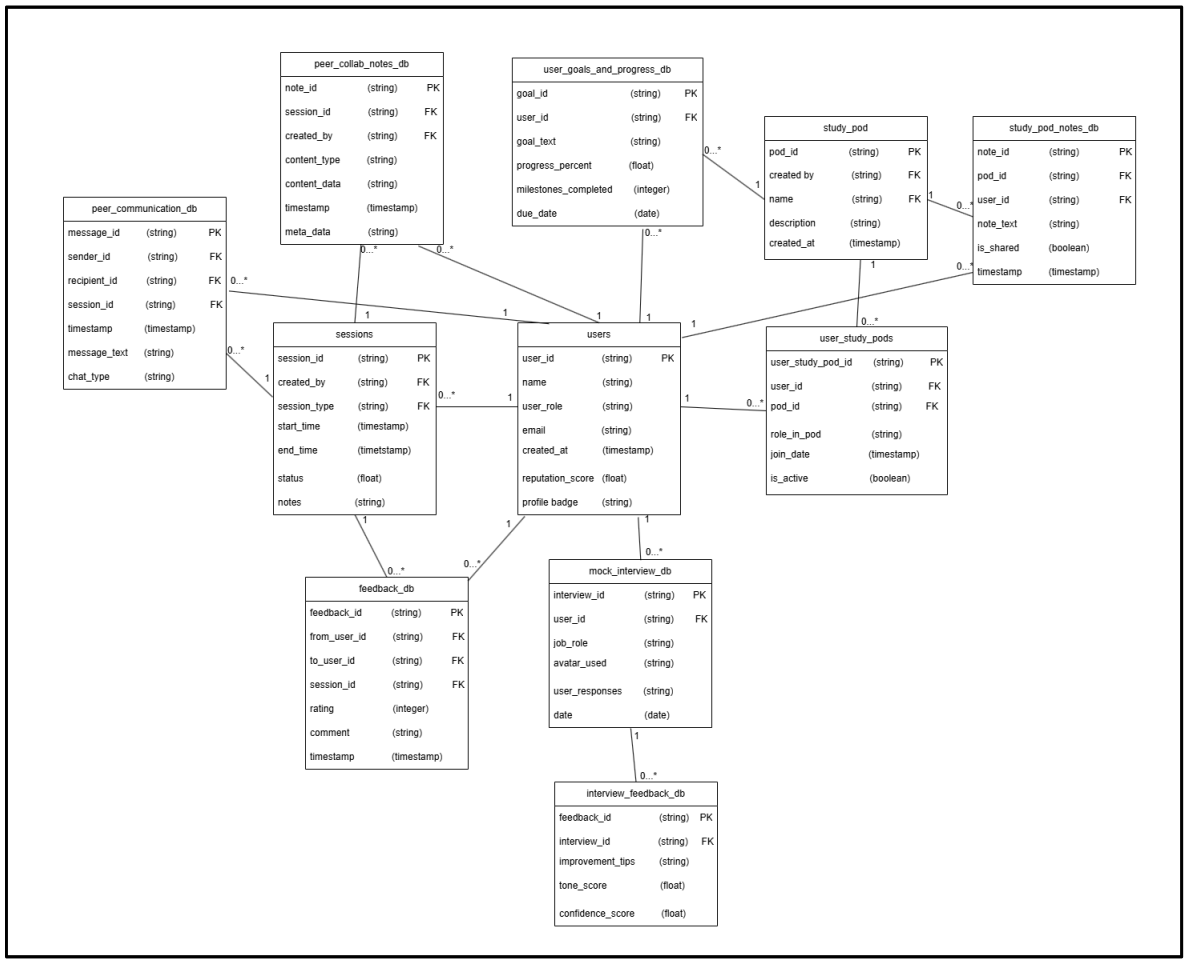
- **Study Pod Notes Database:** This stores all notes taken by users during Study Pod sessions, including personal notes and shared ones. It helps users review what was discussed or learned in past sessions at any time.

- **Feedback Database:** This collects all the star ratings and comments users give each other after Peer Link or Study Pod sessions. It is used to calculate average scores, update reputation, and show feedback on user profiles.

- **Mock Interview Database:** This keeps a record of all mock interview sessions done with the AI avatars in Career Companion. It stores details like the chosen job role, the avatar used, and the user's answers during the interview.

- **Interview Feedback Database:** This database saves the AI's feedback after each interview session. It includes tips on how users can improve their answers, tone, and confidence based on how they performed.

ERD Diagram



This ERD diagram represents a structured system that is designed to support peer collaboration and learning, progress tracking, and many more for SkillSync. At the center of this system is the 'users' table, which stores the essential details of the users such as name, email, reputation and role. A user can have zero or more goals within the 'user_goals_and_progress_db', which records progress status, goal information and milestones. The users are also able to generate multiple peer collaboration notes in the 'peer_collab_notes_db', where each is linked to a session and storing content information and time records.

The 'peer_communication_db' manages message exchanges, where each user can send and receive multiple messages. Each message includes identifiers for both the sender and recipient and may also reference a specific session. The 'session' table records scheduled activities, with each session being created by one user, though a single user can create several sessions. Sessions can also include related notes and feedback.

Feedback is recorded in the 'feedbacked', in which a user can submit and receive multiple feedback records, where each is associated with a session. Mock interviews are handled in 'mock_interview_db', where a user can attempt zero or more interviews, and each of the interviews might have detailed feedback in the 'interview_feedback_db', that is linked to both the interview and feedback record.

Study groups are managed using the 'study_pod' table. Each user has the ability to create multiple study pods, and participation is tracked through the 'user_study_pods' table, establishing a many-to-many user-to-pod relationship. Users can also contribute pod-related notes, which are stored in 'study_pod_notes_db' and are associated with both the user and the corresponding pod.

Overall, this ERD diagram reflects the system with clearly defined associations that ensures all the user activities such as sessions, goals, communication, collaboration and feedback are well lined and easy to follow across the SkillSync platform.

- **Business logic modules**

The business logic modules are the main parts of the system that convert user actions into system responses. They manage how the system's features operate by following specific rules and processes. Each module takes user input, processes the information, and updates the system to reflect changes visible to users. Together, they enable smooth interaction between the user interface and the data layer, maintaining consistency and real-time updates.

1. **Study Pods Module**
    - **Input:** Users create or join study pods, define the topic, set an objective/goal, choose mode of collaboration and deadline.
    - **Process:** The system organizes the goal into smaller, adjustable phases, keeps track of which milestones the user has marked as done, and calculates overall progress.
    - **Output:** Users see their updated progress displayed visually through a progress bar, showing how far along they are toward completing their goal.

2. **Peer Link Module**
    - **Input:** Users post questions, reply to chats, and schedule meetings through the Study Pods.
    - **Process:** The system displays user questions to the Peer Link feed, manages peer-to-peer matching through Study Buddy Match, and disables cameras during meetings while enabling screen sharing.
    - **Output:** Users see real-time chat updates, view their matched study partners, and are able to use a shared whiteboard during meetings.

3. **Peer Feedback System Module**
    - **Input**: After getting help through Peer Link or finishing a study pod session, users rate their partners using a 5-star system based on how clear, prepared, and easy to talk to they were.
    - **Process**: The system calculates the average rating for each user and updates their profile. If someone gets a very high score (like 4.8 to 5 stars), they get a special tag like "Top Mentor."
    - **Output**: Users can see their rating, a Reputation Score, and tags on their profile so others know how good they are to work with.

4. **Career Companion Module**
    - **Input**: Users choose a job role they want to practice for and decide if they want to upload a video of their answers or talk to an AI avatar interviewer.
    - **Process**: The system runs a practice interview using an AI avatar with a unique style. It listens to the user's answers and gives helpful tips.
    - **Output**: Users get feedback on how to sound more confident, clear, and professional, helping them get better at interviews.

- **Licensed components**

Component	Part/Description	Used For	Pre-required Part

HeyGen	AI avatar development platform for creating interviewer personas.	AI Avatar Interview Module	N/A
OpenAI (ChatGPT)	Creates interview questions and feedback dialogues dynamically.	AI dialogue generation	N/A
Assembly AI/Hugging Face	Evaluates transcribed interviews for tone, understandability, and confidence.	Sentiment & feedback scoring	Audio from the interview
FullCalendar.js	Displays an interactive calendar for study pod timelines and scheduling	Study Pod scheduling	N/A
Bubble.io	Used to build and host the SkillSync web application with AI plugin support.	Hosting SkillSync MVP	N/A

- **APIs and external connections**
  - **Authentication:** Handles user sign-up, login, and role-based access (Mentor, Learner, Admin). It secures sessions across all modules like PeerLink, Study Pods, and Career Companion. **Tools:** Firebase authentication.
  - **Study pod and progress tracking:** Used for creating and managing study pods, storing user goals, progress milestones, and timelines. Communicates with MongoDB to store data and Firebase for user state. **Tools**: Express.js (Node.js backend).
  - **Real-time collaboration: WebRTC** - supports live video/audio features in study pods and peer link. **WebSocket Integration** - Enables instant messaging, shared whiteboards, and real-time updates. **Google Meet** - provide enhanced call functionality where needed.

- **Peer communication and matching**: **Study buddy matching** - uses a rule-based algorithm to match users with similar goals or topics. **Quick help chat** - provides real-time text support and screen sharing during sessions.

- **AI avatar interview: HeyGen** – Generates speaking avatars for interview simulations. **OpenAI API** – Powers interview questions and dialogue generation. **AssemblyAI** – Transcribes user audio answers into text. **Hugging Face Sentiment Analysis** – Analyses tone, clarity, and confidence to give interview feedback.

- Note and whiteboard: **Firebase Realtime Database** – Stores collaborative notes and whiteboard updates. **Excalidraw** – Enables shared whiteboard editing across users.

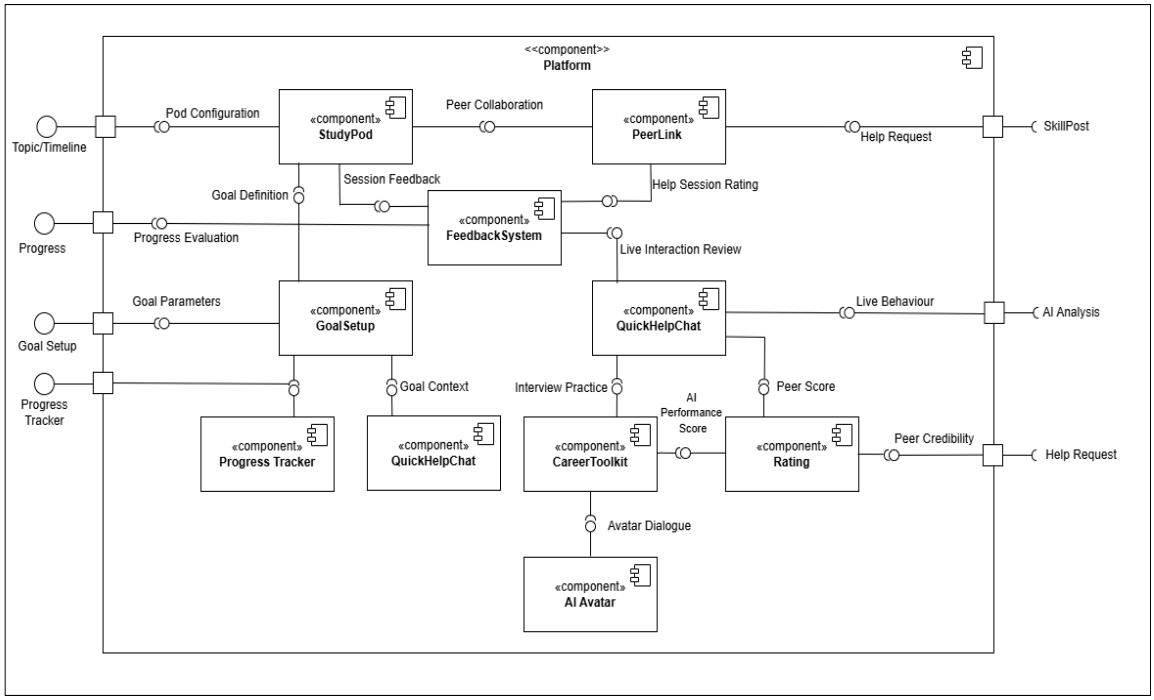
- **Feedback and rating API:** Captures user ratings after Study Pods and PeerLink sessions. Calculates average scores and updates reputation badges. **Tools:** custom express API + MongoDB

- **Notification and reminder**: Sends email or in-app notifications about session times, pod activity, upcoming deadlines, and feedback reminders. **Tool:** n8n (Automation Platform)

- **AI integration**: **OpenAI & Hugging Face** – For generating responses, analyzing user language and tone. **Recorder.js** – Captures audio/video of interviews for user review. **Firebase Storage** – Stores video files and interview session recordings.

- **Gateway, REST, Database APIs: Gateway API**– directs requests, validates authentication, and routes them to appropriate services. **REST API**– Provides communication between the client and server for actions like creating pods, updating the goals, submitting feedback by users. **Database API**– Communicates with MongoDB to perform CRUD operations on users, pods, goals, feedback, and session data.

- **Software component diagram**



This system component diagram shows the structure of the Collaborative Learning Platform that facilitates such aspects as study pods, peer support, and artificial intelligence-powered career tools. Each system component is deliberately created to serve a specific purpose—such as setting goals, tracking progress, interacting with fellow learners, or receiving feedback.

At the center is the Platform component, to which all the other services are bound. It manages interactions between components like StudyPod, PeerLink, QuickHelpChat, CareerToolkit, and FeedbackSystem, providing an integrated experience for users through learning, collaboration, and preparation tools.

Core Functional Elements

- StudyPod: Responsible for creation, setup, and joining grouped study pods. Students define learning subject matter and duration through the Topic/Timeline interface. It is connected with goal setup, feedback, and peer collaboration modules for customized study routes and shared milestones.

- GoalSetup: Enables users to define structured, modular goals of learning. This component processes Goal Setup input and returns Goal Parameters to relevant downstream services like QuickHelpChat for context-based interaction.

- ProgressTracker: It handles real-time progress updates, visual feedback (e.g., bars or milestones), and periodic work. It processes input via the Progress Tracker port and interacts with the Goal and Feedback subsystems.

- FeedbackSystem: Collects and processes structured feedback from peer assistance sessions, study pods, and AI mock interviews. This data is then sent to the Rating component to compute user credibility scores and reputation levels.

Social Collaboration

- PeerLink: Includes the PeerLink system, which supports posting help requests (SkillPost), browsing by topic, and collaboration by whiteboards, code, or text. It interfaces directly with both FeedbackSystem and StudyPod for ratings and cross-module support.

- QuickHelpChat: Supports real-time assistance by text/video chat, typically in spurts of brief help. Enables context-sensitive help by retrieving user-specified objectives and offers interaction feedback to be rated.

Career Companion & AI Feedback

- CareerToolkit: Provides mock interview simulations from user-selected roles. Provides feedback as performance metrics and sends information to the Rating system for tracking credibility.

- AI Avatar: It mimics interviewers of various personality types and provides interactive feedback. It is highly integrated with CareerToolkit in order to be more realistic and generate Avatar Dialogue.

- Rating: Aggregate scores from feedback, live sessions, and mock interviews. Calculates the Peer Score and Peer Credibility, which influence matchmaking and user trust indicators.

Interface Connectivity

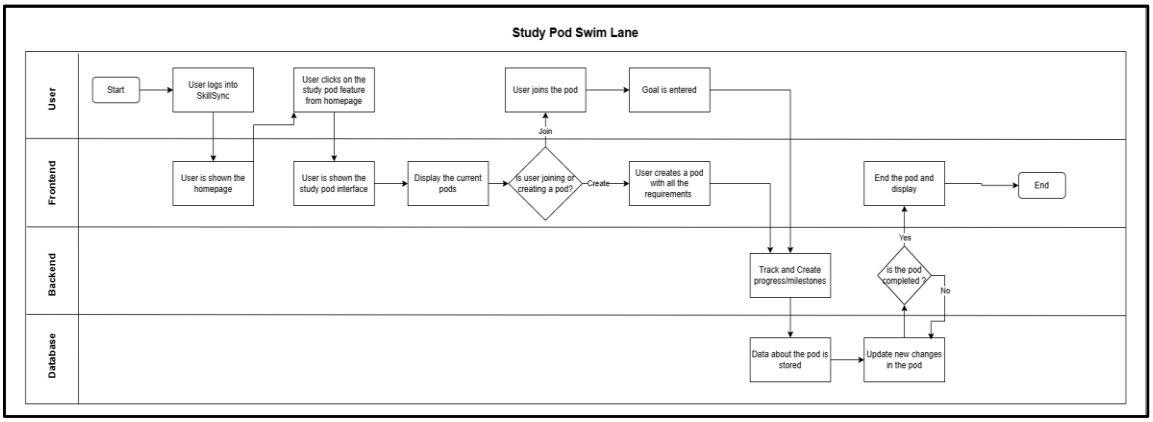
- The left of the diagram marks significant user-driven interfaces such as: Topic/Timeline, Progress, Goal Setup, Progress Tracker.

- These ports are raw user input or interaction that pipe into backend processing blocks.

- The right-hand side provides interfaces that are needed, where services like AIAnalysis, SkillPost, and Help Request are imported by units needing external input or third-party services.

Transactions and User Flows

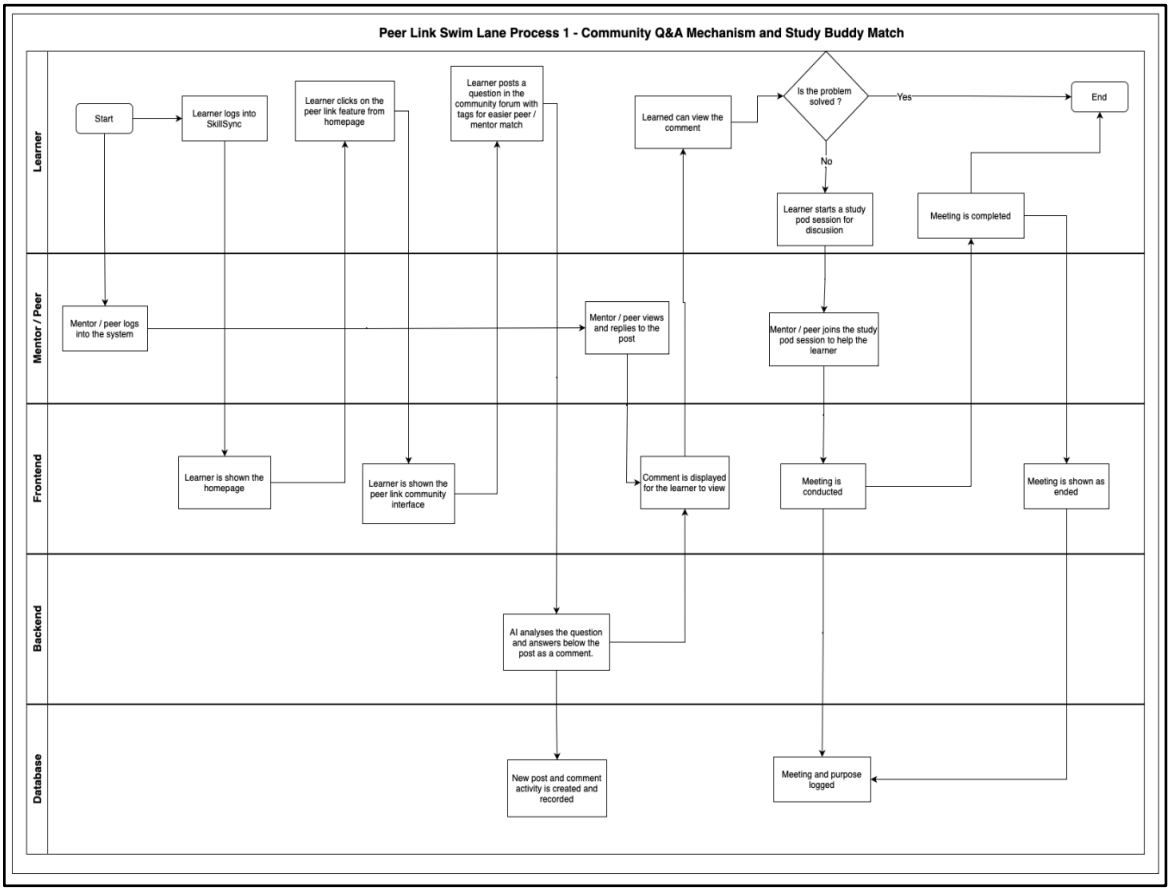
- **Study Pod:**



In this swim lane diagram, the lane is divided into 4: Users, Frontend (system), Backend (system) and Database (system). The flow of the process starts with the user logging in to the application and the frontend displays the homepage. The user selects the study pod feature from the dashboard and the user is directed to a study pod landing page. The user is given the choice to create or join an existing pod. If the user is creating a pod, they will be asked to fill out the requirements needed to start a pod. If the user is joining, they need to enter their specific goal that they wish to achieve by the time the pod course is over.

After creating or joining the pod, the goal is broken down to phases for easier progress tracking by the system and the data created so far about the pod is stored in a database. The new changes are updated in the database and system checks if the pod course has been completed or not. When complete, the pod is ended and the process is done, if it's not over, the system keeps updating up until the last and ends the process

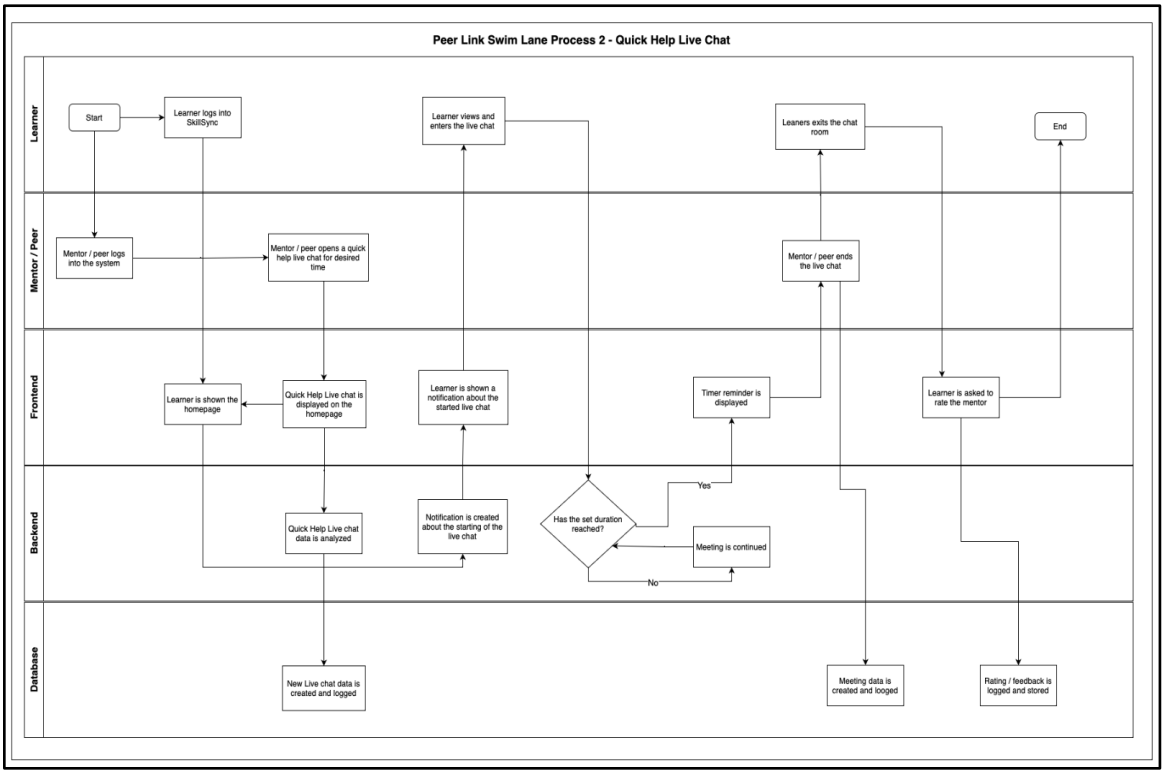
- **Peer Link:**



In this swim lane diagram the lanes are divided into five: Learner, Mentor/Peer, Frontend (system), Backend (system) and Database (system). The process begins when both the learner and the mentor log into the application. After logging in, the frontend (user interface) displays the homepage for both the users. The learner chooses the PeerLink option from the homepage and is directed to the peer community section. The learner then decides to raise a question in the Community Q&A section, with tags to receive a response to their question in an ideal and understandable format. The backend analyzes the question utilizing AI and then generates a relevant or supportive comment promptly, so the user doesn't need to wait for a response. This comment is then stored in the database and shown on the frontend for all users to view it.

The mentor or peer sees the post and replies to it, and the learner can go through the responses. At this stage, the system verifies if the learner acknowledges that the issue has been resolved, if yes, then the process ends there. If not, the learner opts to start a study pod for a detailed discussion. Then mentor or peer joins the study pod session. The session is then conducted, and after it ends, the system records the meeting

details and session purpose into the database. The frontend shows a message which indicates that the session has ended. This finalizes the end of the process, supporting learners at both focused group sessions and general discussions.

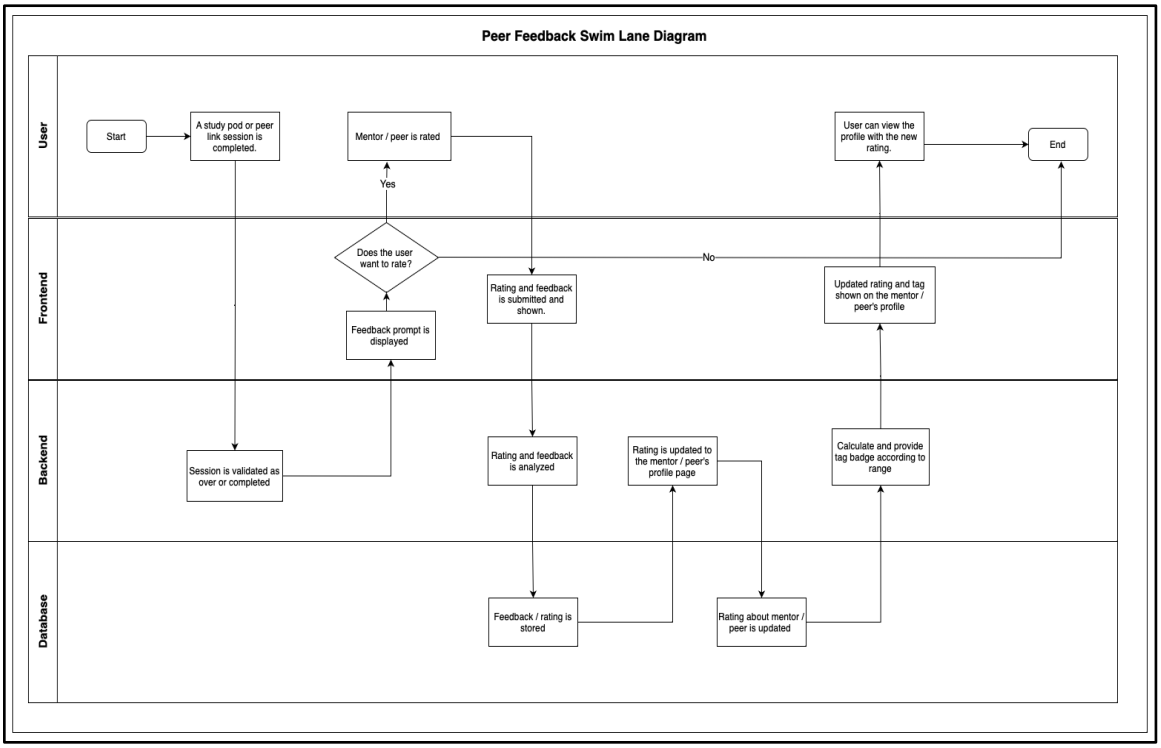


In this swim lane diagram the lanes are divided into five: Learner, Mentor/Peer, Frontend (system), Backend (system) and Database (system). The process begins when both the learner and the mentor log into the application. After logging in, the frontend (user interface) displays the homepage for both the users. The mentor marks themselves as 'available' for quick help, and this can be viewed through the homepage or peer link services. The learner then views and joins the live chat. Once the learner accesses the chat, they can converse with the mentor regarding any queries that could enhance their learning. At that moment, the backend starts analyzing the live chat session and prepares a notification about the rating prompt that will appear at the end of the session.

The system launches a timer to track the duration of the session. If the allocated duration runs out, a notification is displayed. If not, the session continues. The mentor has the option to end the session at any time, after which prompts the learner to leave the chat room. Once the chat concludes, the learner is requested to provide a rating for the mentor. This feedback is saved in the database along with the session

recordings. The system monitors this data for quality control and future insights, thereby marking the end of the help process.

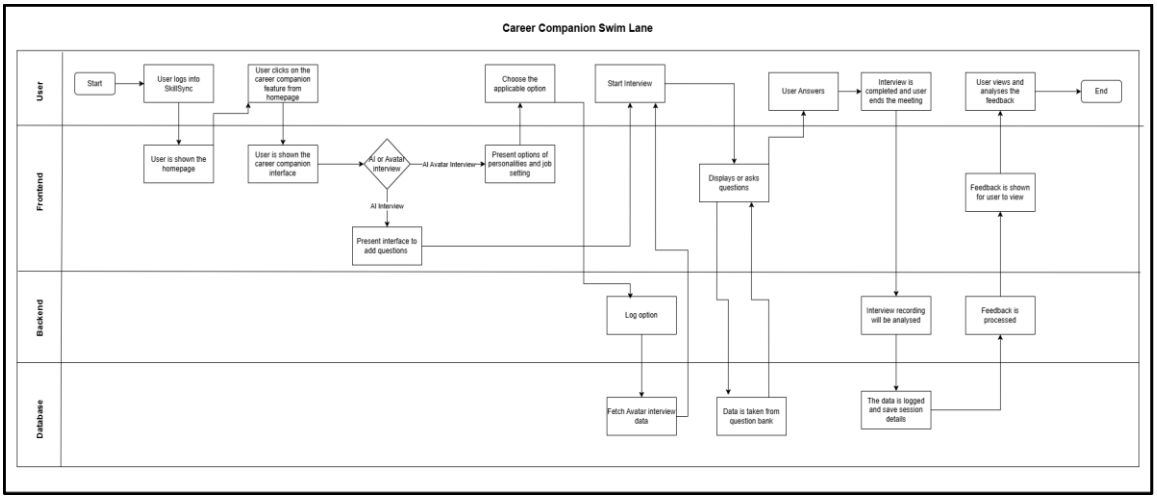
- **Peer Feedback:**



In this swim lane diagram the lanes are divided into four: User, Frontend (system), Backend (system) and Database (system). The process begins when the user completes either a Peer link or Study Pod session. Once the session concludes, the backend system confirms that the session is flagged as completed. After this, the frontend system shows a feedback prompt asking whether the user would like to rate their peer or mentor. If the user opts to not give any feedback, then the process terminates immediately. However, if the user decides to provide feedback, they go ahead to submit a rating, and type in feedback, which is then shown on the interface.

This feedback is sent to the backend system, where both the rating and feedback are analyzed for relevance and quality. Then , the backend saves the rating and feedback into the database. It then updates the rating data for the associated mentor or peer in the database. After this, the backend computes the average rating and allocates a tag badge such as “Top Mentor” according to the predefined rating criteria. These updates, consisting of both tag badge and numerical rating are then pushed to the frontend system, where they display it on the peer/mentor’s profile page. In the end, the user can see the updated profile, and then the process ends.

- **Career Companion:**

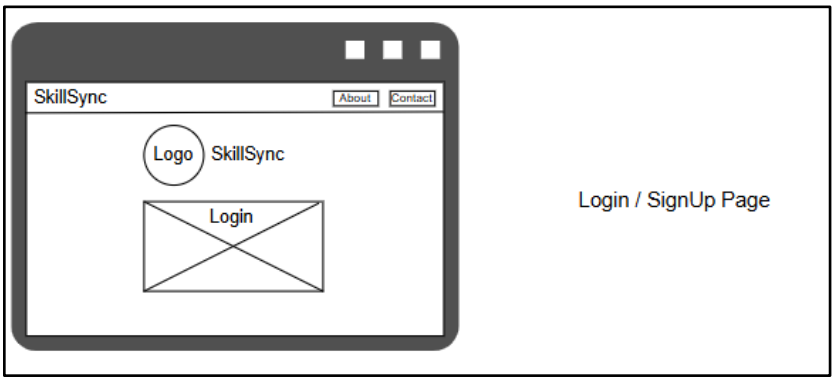


In this swim lane diagram, the lane is divided into 4: Users, Frontend, Backend and Database. The flow of the process starts when a user login to the system and views the homepage of the website. The user selects the career companion feature from the dashboard and the user is directed to its respective landing page. The user is given the option to have an AI interview where the system asks questions regarding the job and position chosen by the user or an AI Avatar interview where they get a feel of a real world interview with different personalities chosen by the user along with job type and position.

If the user chooses the AI Avatar interview, the character option is logged and data for AI Avatar is fetched from the database. The interview is started by the user according to the option they chose, and the questions are asked from a question bank from the database. The user answers the questions and completes them by ending the interview. The recording of the interview is saved as a session in the database and the feedback is processed for the user to review and practice again. This also marks the end of the process.

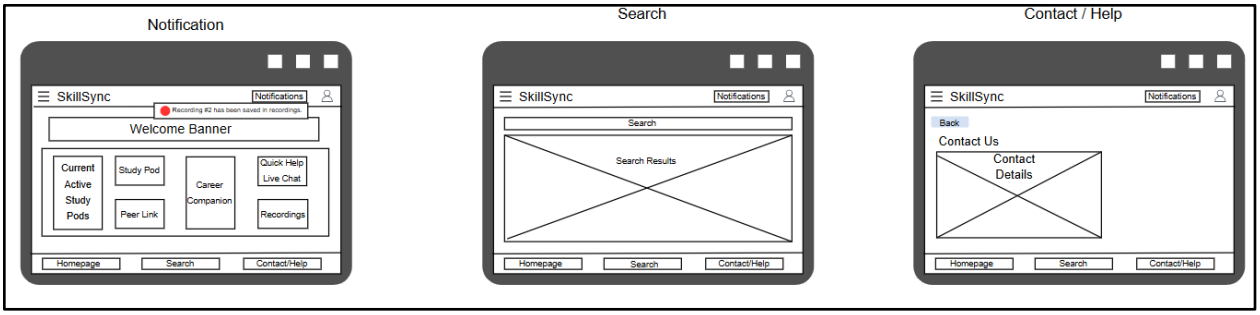
User Interface Design

Log In / Sign Up



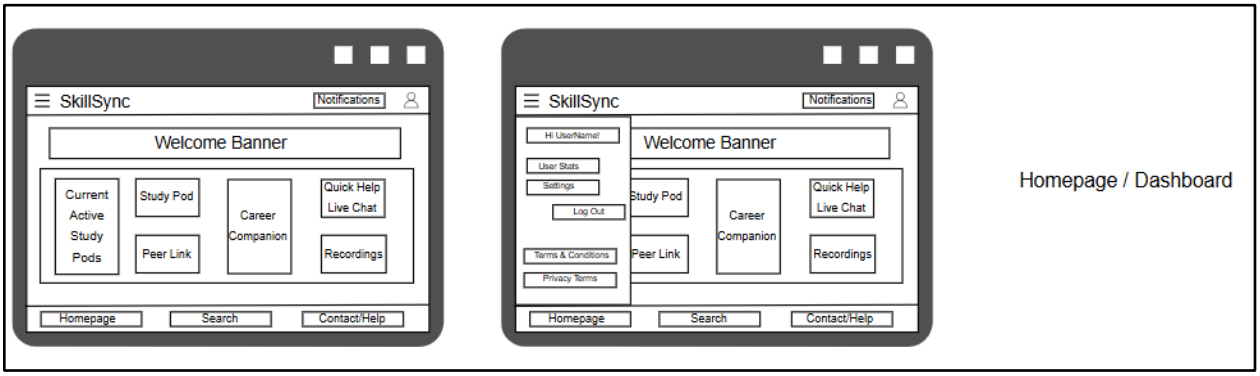
The wireframe above shows the landing page of the website application SkillSync. In this page, the user is prompted to login or signup to use the features of the application. The user can also contact or learn about the application more by navigating and clicking the buttons on the top right corner.

Notification, Search and Contact / Help



These designs show how the potential placements of the three features which are notifications, search and contact/help would be designed. The notification is placed on the top right corner for easier visibility of alters and it can be viewed by clicking on it. The search engine can be accessed from the bottom to search about anything from the website for example an interview recording. The contact/help page can be accessed from the bottom panel by the user to solve any issues or bugs present in the application.

Homepage / Dashboard



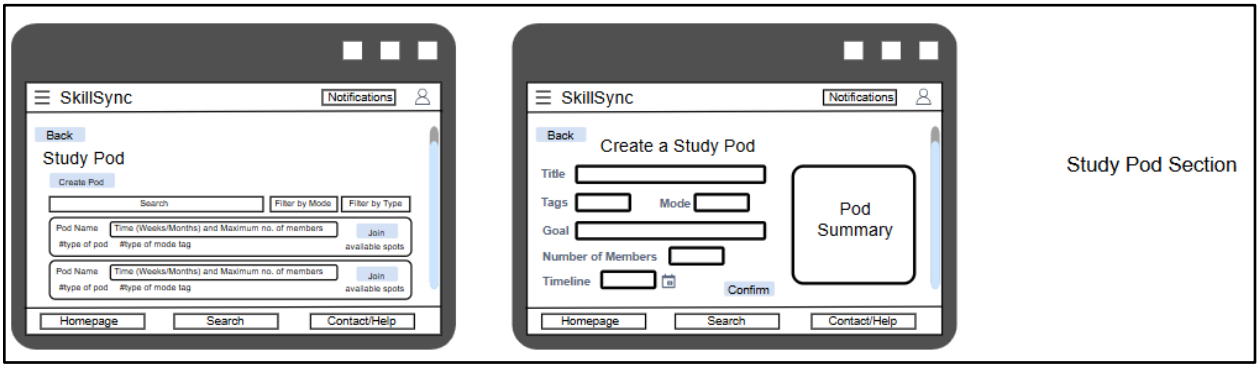
The wireframe design above shows the homepage which the user is navigated to after logging in. In this page, the user can check their profile by clicking on the user shaped icon on the top right corner and their notifications which is situated right next to the profile icon. The user is welcomed with a banner and sees a dashboard that connects them to all the features of the application.

In the dashboard, the user can see their current active pods which acts as a reminder, the study pod feature which will be accessible once they click on Study Pod, the peer link feature which will be accessible once they click on Peer Link, the career companion feature which will be accessible once they click on Career Companion, Quick Help Live Chat which lets user access it faster and Recording which lets them view their interview recordings with ease.

Next to the name SkillSync, the user can access the side menu which includes features such as user statistics such as their activity, settings and log out. It also includes Privacy terms as well as terms and conditions about the application which can be accessed easily by the user.

The lower panel includes homepage which lets the user navigate to the homepage faster, search which lets the user find their resources or end target with ease and contact/help which can be used by the user to get their technical or website related problems solved faster by contacting SkillSync.

Study Pod

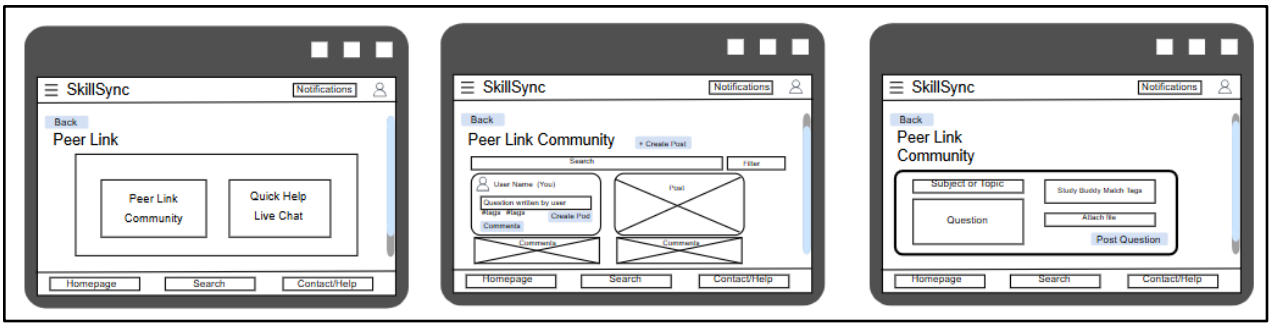


This user will be able to access the study pod section by clicking on Study Pod from the homepage/dashboard. In this section the user can create or join the pod. The user is taken to a page where they can view existing pods started by other users. Each pod will first have information about it such as the name, tags such as programming, doubts discussion, exam preparation, this makes it easier for users to filter and find their desired pods.

The pods will include the mode of learning such as online, independent, group or mixes. The timeline and the maximum number of members will also be mentioned; this will aid the users in navigating and join the correct pod.

To create a pod, the user needs to click on the button “Create Pod” which takes them to a form like page where all the requirements need to be filled. Once all is filled, the pod summary shows in the side for the user to see and make changes to the requirements. Once done, the user can click on “Confirm” which will proceed to create the pod.

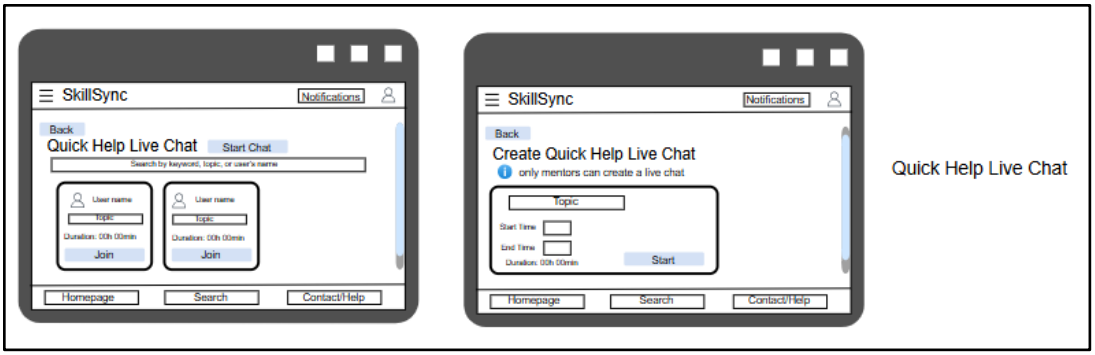
Peer Link



The Peer Link dashboard page provides the user with access to two features: Peer Link Community and Quick Help Live Chat. When a user navigates to the Peer Link Community page, they can ask a question by providing the subject or topic, writing their question, and adding relevant study buddy tags that reflect the kind of help they need. They also have the option to attach a file. Once posted, the question becomes visible to the community, and

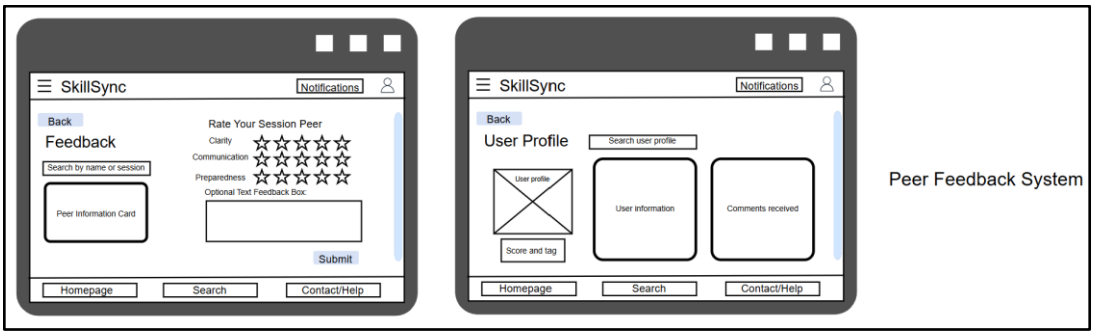
other users can respond with helpful answers. If a user finds a response particularly useful and wants to explore the topic further, they can schedule and create a pod session with the responder for a more in-depth discussion.

Quick Help Live Chat



This feature is part of the peer link feature, where mentors specifically can only start a live chat. The mentor can set the start and end time, and the duration of the chat is displayed below. A topic is also added for easier understanding so that learners can join the correct meeting to solve their queries. The search is done by topic or user name for easier detection. Quick help live chat can be accessed through peer link from the dashboard in the homepage or directly from the dashboard itself for accessibility purposes.

Peer Feedback



On the Feedback page, users can rate someone by either searching for the user by name or finding them through a past session they've shared. After selecting the user, the peer's information card will appear, showing key details about them. Users can then rate them on clarity, communication, and preparedness using a five-star system, with the option to leave written feedback before submitting the review.

Career Companion

The Career Companion dashboard page gives users access to three sections: Job Role Interview, Job Role Interview with AI Avatar, and a section to view their Recordings and feedback.

- **Job Role Interview**
  When the user accesses the Job Role Interview page, they can select the role they want to practice for and click the "Start Interview" button to begin. During the interview, they will view and respond to questions related to the selected role. Once completed, the recording and feedback will be automatically saved in the recordings section.

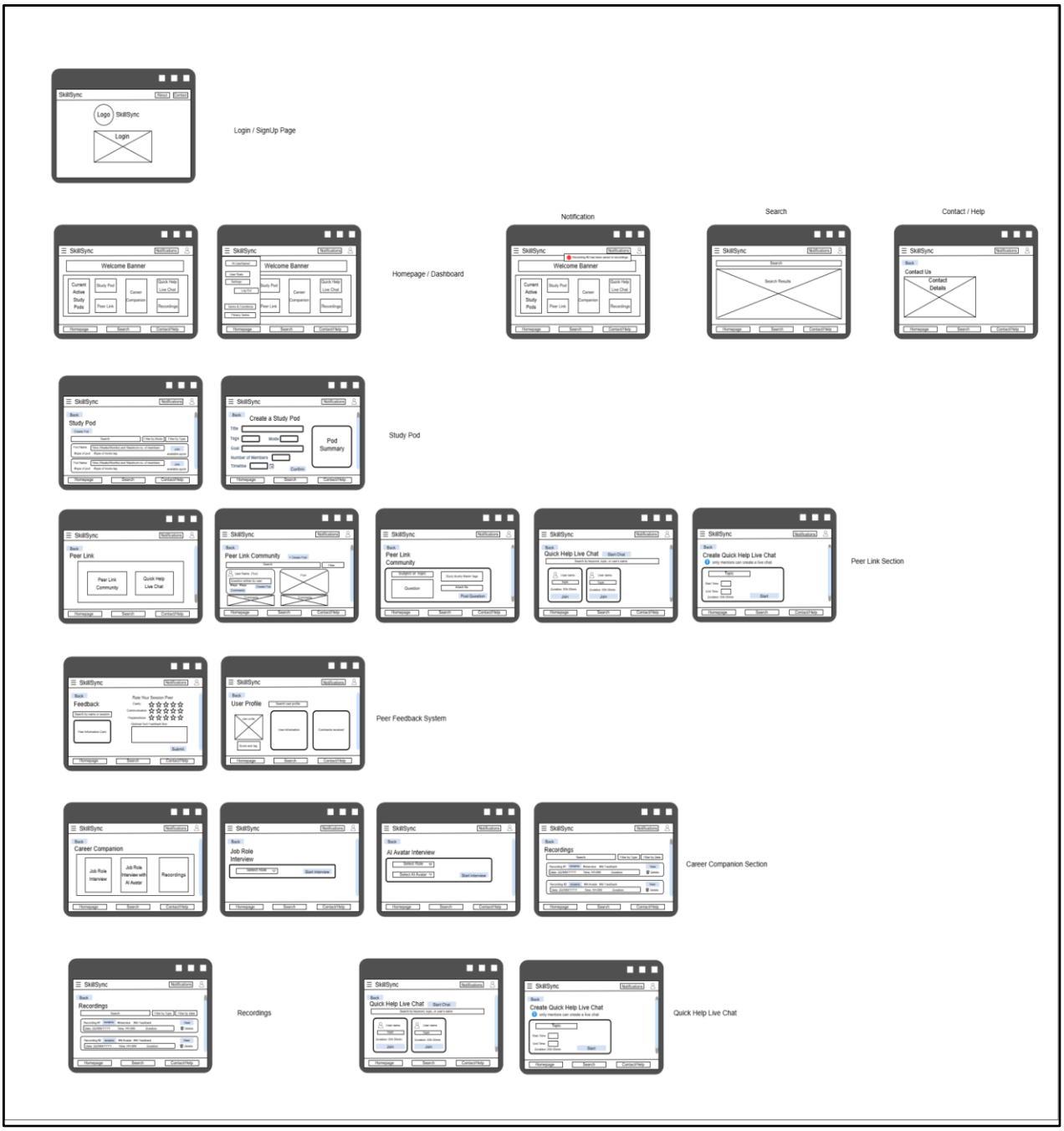
- **AI Avatar Interview**
  When the user accesses the AI Avatar Interview page, they can select the role they want to practice for and choose the AI Avatar interviewer they want to practice with. After that, they click the "Start Interview" button to begin. During the interview, they will view and respond to questions related to the chosen role while interacting with their selected AI Avatar. Once finished, the recording and feedback are automatically saved in the recordings section.

Recordings

This feature is part of the career companion where users can view their interview recordings. The recordings are saved with the title, tags, date, duration and time. The titles can be

renamed so that the users can access the feedback easily. The user can delete the file if needed but a confirmation will be asked for error recovery purposes. Recordings can be accessed through career companion from the dashboard in the homepage or directly from the dashboard itself, this was done for easier accessibility.

The whole wireframe of the user interface:



Deployment and Environment

The SkillSync platform will be deployed as a web-based application hosted in the cloud, which will be accessible worldwide through standard web-browsers. The system's backend will be installed and managed in a cloud environment, either Amazon Web Services (AWS) or Google Cloud Platform (GCP). This guarantees strong system stability, flexible system scalability, and load balancing that is managed. The applications operate on the browser interface, so the local installation on the client-side is unnecessary.

From the server's perspective, the architecture is designed to handle backend functions successfully. The server-side configuration includes Node.js and Express.js installations to run the backend services, MongoDB and Firebase for database and storage for database and storage management, and integrated APIs for AI services such as OpenAI, HeyGen, AssemblyAI, and Hugging Face. The services are designed to communicate with one another within the cloud infrastructure, with additional components like n8n for automation and Firebase Authentication for secure login session management.

From the client's perspective, the platform runs entirely through a web browser, eliminating the need for an app download or local installation. To access the platform, users simply need a device which meets the basic specifications, such as: 4GB of memory, a dual-core processor of 1.5 GHz or above, a minimum of 2GB of available storage space, and lastly, a reliable internet connection of 10 Mbps or more. This system supports browsers such as Safari, Chrome, and Firefox, as well as accessible on mobile operating systems such as iOS and Android.

The System needs to be executed in three primary environments:

1. Development Environment – Hosted locally or in staging on the cloud, with mock data and emulators for testing in safety.
2. Staging Environment – A standalone pre-production cloud environment used for internal QA, update testing, and checking configurations.
3. Production Environment – The production SkillSync system is ready for end-users, running on secure cloud servers with monitoring, load balancing, and full AI service integration.

In short, the SkillSync platform is cloud-oriented, deployed and maintained on cloud-hosted servers, which are accessed via internet browsers, and is designed to function across diverse software and hardware environments without requiring a local setup.

Security Considerations

Security is a critical aspect of the SkillSync platform, given its dependence on AI-based learning features, live collaboration, and handling of sensitive information of the users such as the recordings of interviews and personal profiles. Authentication is managed safely through Firebase Authentication, which implements management of user sessions and role-based access rights. This guarantees that only verified users such as mentors, advisors, or peers, are permitted to access the key features including feedback submission, enrolling study pods, or participating in mock interview sessions. All confidential information, such as chat transcripts, objectives, and video materials, is safely maintained in MongoDB and Firebase storage, both of which support encryption for both data transfer, and storage to avoid unapproved access and leakage of data.

To further enhance security, the platform integrates multiple additional security measures. Firebase Security Rules are applied to strictly manage write permissions and data read, and management of token-based systems with timeouts that are automatic to prevent the risk of session use that is unauthorized. To secure all the data exchanges, HTTPS encryption is implemented across the whole application. For interactions between peers, to encourage accountability and reduce misuse, a feedback-based credibility system is utilized. Understanding that potential risks such as AI misjudgments, spam, or misconduct in community forums may still emerge, the developers apply regular inspection tools and testing strategies including Postman for API validation and Firebase Security testing for detecting and then resolving vulnerabilities in advance. To mitigate these risks, SkillSync implements Firebase Security Rules to restrict data access permissions, utilizes sessions that are token-based with timeouts to secure user activity, and actively monitors the quality of peer feedback. This multi-layered security framework ensures that SkillSync continues to remain reliable, safe and in line with the best practices of the modern web security standards.

Risks and Mitigation Strategies

Risk	Description	Impact	Likelihood	Mitigation Strategy
Low User Engagement	Users may not use the app to its full potential or get overwhelmed, leading to underutilization of	Medium	Medium	- Use push notifications and reminders. - Provide new users with helpful guides or step-by-step

	key features such as Study Pods, PeerLink, and Career Companion.			instructions to highlight feature benefits and help them use key features effectively.
Mismatched Study Buddy Pairs	Users may be matched with incompatible peers (different skill levels, time zones, or communication styles).	Medium	Medium	- Improve matching algorithms using interests, ratings, and preferences. - Allow filters and rematching options.
Poor Collaboration Experience	Real-time tools like the whiteboard or live chat may lag or crash, affecting productivity.	High	Medium	- Use scalable, reliable third-party services (e.g., Firebase, WebRTC). - Test under load and optimize front-end performance. - Implement fallback options (e.g., autosave, offline mode).
AI Feedback Reliability (Career Companion)	AI may give inaccurate or misleading feedback.	Medium	Medium	- Train AI on high-quality data. - Continuously improve AI models with user feedback. - Allow users to flag inaccurate feedback. - Offer disclaimers that feedback is advisory, not absolute.
Privacy and Security	Risk of information exposure or misuse	High	Medium	- Encrypt data at rest and in transit.

Concerns	of collaboration tools.			- Implement access control and role-based permissions. - Comply with relevant data protection laws to safeguard user privacy.

Out-of-scope

The initial release of SkillSync focuses on basic collaborative learning and career-readiness capabilities, with several advanced features intentionally withheld to reduce scope and make it feasible. No particular mobile application will be developed at this stage, as the platform is optimized for web use alone. Offline viewing of sessions, study pods, or interview content is not within scope. The system won't deliver accredited certifications or support enterprise customers. Multilingual support is deferred for future development, while AI-driven avatars will be applied solely for application within the career companion module but not study pod or peer link sessions. Additionally, sophisticated moderation tools like automatic content filtering or behavior tracking will be absent from the MVP, where moderation will be community reporting-based.

Team Work Distribution

Report Parts	Student Name	Student ID
Introduction	Sara Ranjbar	7817848
Requirement Summary	Yaxin Wang	7741091
Assumptions & Prerequisites	Yaxin Wang Cadence D'Souza	7741091 7769714
High-Level Design		
High-Level Software Design	Sikha Nair	7810398
Use Case Diagram	Sara Ranjbar	7817848

Activity Diagram	Yaxin Wang	7741091
Integration Issues	Sikha Nair	7810398
Architectural Plan		
Physical Architecture	Sharen Thomas	7700787
Logical Architecture	Cadence D'Souza Yaxin Wang	7769714 7741091
Application Modules		
Core Components	Sikha Nair Sharen Thomas	7810398 7700787
User Interfaces & Layers	Sikha Nair Sara Ranjbar	7810398 7817848
Database Layer	Sharen Thomas Yaxin Wang	7700787 7741091
Business Logic Modules	Sara Ranjbar Yaxin Wang	7817848 7741091
Licensed Components	Cadence D'Souza	7769714
APIs & External Connections	Cadence D'Souza Yaxin Wang	7769714 7741091
Software Component Diagram	Cadence D'Souza	7769714
Transactions & User Flows	Sikha Nair Sharen Thomas	7810398 7700787
User Interface Design	Sikha Nair Sara Ranjbar	7810398 7817848
Deployment & Environment	Sharen Thomas Cadence D'Souza	7700787 7769714
Security Considerations	Sharen Thomas	7700787
Risks & Mitigation Strategies	Sara Ranjbar	7817848
Out-of-Scope	Cadence D'Souza	7769714